DEVPOST LEARN HACKATHON • MASTER PROJECT CATALOG

# Build With AI: Basics — Master Project Dossier

Complete, unshortened compilation of all 9 high-impact proof-of-concept project proposals. Each proposal includes its full problem statement, end-to-end loop, tech architecture, judging alignment, and demo video script.

**Hackathon:** learn-ai-basics.devpost.com

**Prize Pool:** $2,500 Cash

**Deadline:** Oct 26, 2026 (5:00 PM EDT)

**Curriculum:** 6-Step Spec-Driven Workflow

## ✓ Hackathon Judging Criteria (33.3% Each)

**1. Design (33.3%)**
A polished, coherent product experience—clean typography, micro-animations, intuitive responsive states, and no broken edges.

**2. Innovation / Idea (33.3%)**
Novelty and creativity. Tackles a real pain point with a clear agentic angle instead of a passive text summarizer.

**3. Presentation (33.3%)**
Demo video under 3 minutes showing the single loop running live on realistic inputs with zero hand-waving or slides.

## 01 BureaucracyPilot — Government Paperwork & Visa Pre-Flight Inspector

PRACTICAL REAL-WORLD 01 **BureaucracyPilot**

CIVIC TECH & ADMINISTRATIVE UTILITY

*"Never get rejected at the counter after waiting 4 hours in line again."*

⚠ PROBLEM SOLVED

Applying for passport renewal, driver's licenses, tourist/work visas, student scholarships, or municipal permits is a nightmare. Official portals are confusing, and people constantly miss 1 stamp, wrong photo dimension, or prerequisite document, wasting entire days in queues.

★ WHY JUDGES LOVE IT

Every human hates government bureaucracy. The utility is immediate and saves real people hours of physical queuing. Looks brilliant on video with progressive checklists and readiness gauges.

→ THE CORE END-TO-END LOOP

1. **Select Process:** User picks or searches a bureaucratic process (e.g., *"Renew Expired Driving License"*, *"Apply for Schengen Visa"*).
2. **Interactive Pre-Flight Checklist:** App generates an exact, plain-English checklist of every required document, fee, and prerequisite.
3. **Document Verification:** User checks off items with smart guidance (e.g., *"Passport expiry < 6 months?"*, *"Does bank statement have official bank seal?"*).
4. **Counter Day Cheat-Sheet:** 1-click printable summary: exact stack order of papers, exact fee amount in cash, and questions the counter officer will ask.

⚙ TECH & DEMO STRATEGY

**Stack:** React / Vite, Structured JSON schema rules, LocalStorage persistence.
**2-Min Video:** Pick "Tourist Visa" → Readiness score starts at 20% (Amber) → Upload/verify bank seal & passport photo → Score hits 100% (Green) → Export Counter Day Cheat-Sheet.

## 02 WarrantyVault — Proof-of-Purchase & Auto-Claim Engine

PRACTICAL REAL-WORLD 02

## WarrantyVault

CONSUMER ELECTRONICS & FINANCIAL PROTECTION

*"Stop losing hundreds of dollars throwing away broken electronics that are still under warranty."*

### ⚠ PROBLEM SOLVED

Laptops, appliances, and gadgets break within 6–12 months. People lose paper receipts (thermal ink fades), don't know when the warranty expires, and writing a formal legal claim to manufacturer RMA teams is intimidating.

### → THE CORE END-TO-END LOOP

1. **Quick Input:** User uploads a photo of a faded paper receipt, invoice PDF, or enters product info.
2. **Auto-Extraction:** AI extracts purchase date, model number, serial info, and detects standard warranty coverage terms.
3. **Countdown Dashboard:** Visual cards with live countdown timers and color-coded status badges (Active, Expiring Soon, Expired).
4. **One-Click Claim Generator:** When a device fails, clicking *"File Claim"* generates a formal, legally grounded warranty replacement demand citing statutory consumer rights.

### ★ WHY JUDGES LOVE IT

Solves a concrete financial loss. Everyone has thrown away broken electronics they paid for. The countdown UI and auto-generated RMA demand letter demo effortlessly in under 90 seconds.

### ⚙ TECH & DEMO STRATEGY

**Stack:** React + Vite, OCR/multimodal parser, Date countdown math, PDF claim generator.
**2-Min Video:** Drop faded receipt → Instantly extracts "Sony Headphones, 42 days left" → Click "Device Broken" → Generates formal RMA claim letter ready to email.

## 03 SmartPantry / ZeroWaste Chef — Grocery & Expiry Economizer

PRACTICAL REAL-WORLD 03 **SmartPantry / ZeroWaste Chef**

COST OF LIVING & FOOD SUSTAINABILITY

*"Turn wilting fridge groceries into gourmet dinners and cut food waste to zero."*

⚠ PROBLEM SOLVED

Due to high food inflation, families spend hundreds on groceries, yet 25% of fresh vegetables and dairy rot in the fridge. People stare at a crowded fridge, feel there's nothing to eat, and order expensive takeout.

→ THE CORE END-TO-END LOOP

1. **Input:** User uploads a photo of their fridge/pantry or pastes a grocery supermarket receipt text.
2. **Expiry Prioritizer:** AI detects perishable items and flags items expiring within 48 hours (e.g., *"Heavy cream expires tomorrow; Spinach is wilting"*).
3. **Zero-Waste Recipe Engine:** Generates 3 recipes using *only* what you currently have (strict constraint: zero extra grocery store trips).
4. **Dollar Waste Saved:** Displays an aggregate running savings counter (e.g., *"You just saved $18.50 by utilizing perishables today"*).

★ WHY JUDGES LOVE IT

Deeply relatable across every socioeconomic background. Clean aesthetic with ingredient tags, step-by-step recipe cards, and an inspiring dollar-savings metric that appeals to judges.

⚙ TECH & DEMO STRATEGY

**Stack:** React + Tailwind/CSS, Vision item tagger, Recipe constraint prompting.
**2-Min Video:** Snap fridge shelf photo → 5 ingredients recognized → Spinach flagged red → Generates "15-Min Creamy Spinach Pasta" → Savings metric: +$18.20.

## 04 CareerLedger — Performance Review & Promotion Evidence Builder

PRACTICAL REAL-WORLD 04 **CareerLedger**

CAREER ADVANCEMENT & WORKPLACE PRODUCTIVITY

*"Stop forgetting your achievements when appraisal time arrives."*

⚠ PROBLEM SOLVED

At annual appraisal time, employees forget 90% of the impactful work they did 6 months prior. They write vague self-assessments (*"I worked hard on the backend"*) and get passed over for promotions because managers only reward quantified proof.

→ THE CORE END-TO-END LOOP

1. **Drop Raw Notes:** User pastes messy, disorganized weekly bullet points, Slack shout-outs, Jira ticket links, or project accomplishments.
2. **Impact Quantifier:** AI translates casual notes into quantified achievements (e.g., *"Upgraded DB" → "Refactored PostgreSQL queries, slashing latency by 42%"*).
3. **STAR Method Formatter:** Structures accomplishments into executive **STAR format** (Situation, Task, Action, Result).
4. **Promotion Pitch Deck / Self-Review Export:** Compiles an appraisal-ready report that the employee can submit directly to their manager.

★ WHY JUDGES LOVE IT

Solves an anxiety-inducing career problem for corporate workers and engineers. The transformation from messy raw notes to executive-level bullet points is an immediate "wow" moment.

⚙ TECH & DEMO STRATEGY

**Stack:** React + Vite, Text transformation engine, STAR framework formatter, Markdown/PDF export.
**2-Min Video:** Paste 4 messy Slack notes → Click "Quantify Impact" → Instant executive STAR bullet points → Export appraisal self-review memo.

## 05 FinePrint Radar — Contract & "Gotcha" Decoder

CORE CONCEPT 01

## FinePrint Radar

CONSUMER PROTECTION & LEGAL AI

*"Never get tricked by sneaky subscription clauses, gym contracts, or SaaS terms again."*

**⚠ PROBLEM SOLVED**

Consumers and small businesses blindly agree to terms containing predatory auto-renewals, mandatory arbitration waivers, automatic 15% annual price hikes, or privacy-invasive data resale clauses.

**→ THE CORE END-TO-END LOOP**

1. **Input:** User pastes text, drops a PDF contract, or picks from built-in sample contracts (Gym agreement, SaaS ToS, Apartment Lease).
2. **Analysis:** AI evaluates clauses against predatory patterns and computes an interactive *Gotcha Index (0–100)* with risk level badges.
3. **Exploration:** Interactive visual risk map highlights offending clauses in red/amber with plain-English translations and financial exposure estimates.
4. **Action:** One-click generation of a customized *Opt-Out / Negotiation Letter* ready to copy or email.

**★ WHY JUDGES LOVE IT**

Universal appeal. Judges instantly understand the value in 20 seconds. Demonstrates practical AI utility with high visual appeal (gauges, interactive clause pins, side-by-side translation).

**⚙ TECH & DEMO STRATEGY**

**Stack:** Vite + Vanilla CSS / React, AI JSON structured extraction, Canvas Gauge.
**2-Min Video:** Open app → Load sample sneaky gym contract → Risk score hits 84/100 (Red) → Click auto-renewal trap clause → Click "Generate Cancellation Email" → Done.

## 06 CritiqueDesk — AI UX & Accessibility Visual Auditor

CORE CONCEPT 02 **CritiqueDesk**

DESIGN TOOLS & WEB ACCESSIBILITY

*"Instant design critique and visual heuristic evaluation for product builders."*

⚠ PROBLEM SOLVED

Indie developers, designers, and students need fast, objective feedback on landing page layouts, accessibility contrast violations, and cognitive clutter before deploying to users.

★ WHY JUDGES LOVE IT

Directly aligns with the "Design" judging criterion by demonstrating design intelligence. The video demo is purely visual and visually captivating.

→ THE CORE END-TO-END LOOP

1. **Input:** User enters a website URL or uploads a UI screenshot.
2. **Audit:** AI scans layout hierarchy, WCAG contrast compliance, call-to-action prominence, and readability metrics.
3. **Visual Overlay:** An interactive canvas displays pinpoint visual pins over the screenshot with severity ratings (High, Medium, Polish).
4. **Fix:** Clicking any pin displays the exact CSS fix or UX recommendation ready to copy.

⚙ TECH & DEMO STRATEGY

**Stack:** HTML5 Canvas / SVG overlay, Vision/Multimodal AI analysis, modern dark-mode workbench.
**2-Min Video:** Upload messy landing page → See audit pins light up → Inspect low-contrast CTA button → Copy recommended CSS fix.

## 07 GitPulse — PR Impact & Architecture Visualizer

CORE CONCEPT 03 **GitPulse**

DEVELOPER PRODUCTIVITY & DEVOPS

*"Turn noisy code diffs into clear visual architecture diagrams and risk briefings."*

⚠ PROBLEM SOLVED

Reviewing large pull requests with 30+ files is overwhelming. Engineers often overlook breaking schema changes, security flaws, or ripple effects on downstream modules.

★ WHY JUDGES LOVE IT

Hackathon judges love developer tools. It provides immense practical value without requiring heavy backend servers or complex dependencies.

→ THE CORE END-TO-END LOOP

1. **Input:** User pastes a GitHub PR diff, commit patch, or git log.
2. **Synthesize:** AI reconstructs affected components, dependency graphs, and breaking change risks.
3. **Visualize:** Interactive node-link dependency diagram highlights modules undergoing high churn or critical API modifications.
4. **Output:** Generates a 30-second bulleted "Reviewer Brief" and an automated QA smoke-test checklist.

⚙ TECH & DEMO STRATEGY

**Stack:** Mermaid.js / Vis.js for interactive topology graphs, Prism.js diff rendering, AI code analysis.
**2-Min Video:** Paste 400-line git diff → Instant visual graph maps affected services → Shows Breaking Change Alert → Exports QA testing checklist.

## 08 SocraticLab — High-Stakes Scenario Simulator

CORE CONCEPT 04

## SocraticLab

INTERACTIVE EDTECH & CAREER SKILLS

*"Master tough conversations through branching AI roleplay and real-time persuasion feedback."*

### ⚠ PROBLEM SOLVED

Practicing high-stakes dialogues (asking for a promotion, handling angry customers, pitching skeptical investors) is impossible to do effectively alone without active feedback.

### ★ WHY JUDGES LOVE IT

Showcases sophisticated agent interaction rather than passive text generation. The live changing radar creates high dramatic engagement on video.

### → THE CORE END-TO-END LOOP

1. **Scenario:** User chooses a scenario and difficulty level (e.g., *"Skeptical CFO on Budget Increase"*).
2. **Interactive Dialogue:** Dynamic conversational simulation with an AI agent holding distinct behavioral traits.
3. **Live Feedback:** A real-time *Persuasion & Emotional Tension Radar* reacts dynamically after each exchange.
4. **Debrief:** Post-session scorecard rating tactical choices, empathy, and objection handling with replays.

### ⚙ TECH & DEMO STRATEGY

**Stack:** Reactive chat UI, Chart.js / Radar Chart for sentiment, dynamic prompt state machine.
**2-Min Video:** Start salary negotiation → Make counter-offer → Live radar scores jump → Close deal → Review tactical feedback scorecard.

## 09 DataStory — Instant Data Narrative & Chart Engine

CORE CONCEPT 05 **DataStory**

BUSINESS INTELLIGENCE & ANALYTICS

*"Drop any messy CSV/JSON and get an executive interactive data story in 5 seconds."*

⚠ PROBLEM SOLVED

Non-technical operators and founders waste hours wrestling with spreadsheet pivot tables and BI dashboards when all they need is a 3-bullet insight summary and 2 clean charts.

→ THE CORE END-TO-END LOOP

1. **Drop Data:** User drags and drops any messy CSV or JSON file.
2. **Auto-Profiling:** Client-side parsing identifies numerical columns, dates, categories, and null values.
3. **Story Generation:** AI detects highest-impact trends, outliers, and correlations, generating 3 responsive charts.
4. **Executive Brief:** Automatically compiles an *"Executive Insights Memo"* with bulleted action recommendations and exportable cards.

★ WHY JUDGES LOVE IT

Unbeatable before-and-after demonstration. The transformation from an ugly text grid into sleek, interactive charts and bullet summaries is instant and visually punchy.

⚙ TECH & DEMO STRATEGY

**Stack:** PapaParse, ApexCharts / Chart.js, Glassmorphic analytics cards.
**2-Min Video:** Drag messy sales CSV → 3 seconds later: 3 interactive charts appear → Key outlier spotlighted → Click export executive brief.

## ▦ Complete Master Strategic Comparison Matrix (All 9 Ideas)

| PROJECT CONCEPT | DOMAIN CATEGORY | VISUAL "WOW" | BUILD TIME | KEY DIFFERENTIATOR & WINNING HOOK |
|---|---|---|---|---|
| **01. BureaucracyPilot** Top Pick | Civic & Legal Admin | ★★★★★ 9.7 | ✓ ~3 Hours | Universal pain; counter readiness gauge & cheat-sheet |
| **02. WarrantyVault** Top Pick | Consumer Electronics | ★★★★☆ 9.4 | ✓ ~2–3 Hours | Receipt OCR & 1-click legal RMA warranty claim |
| **03. SmartPantry** | Household & Food Waste | ★★★★☆ 9.2 | ✓ ~3 Hours | Zero extra grocery run; dollar waste saved tracker |
| **04. CareerLedger** | Career & Workplace | ★★★★☆ 8.9 | ✓ ~2 Hours | Messy raw notes → executive STAR appraisal bullets |
| **05. FinePrint Radar** | Consumer Legal AI | ★★★★★ 9.8 | ✓ ~3 Hours | Gotcha index (0–100) & redline contract diff view |
| **06. CritiqueDesk** | UI/UX Design Auditor | ★★★★★ 9.5 | ✓ ~3–4 Hours | Visual pin canvas overlay directly on UI screenshots |
| **07. GitPulse** | DevTools / CI Review | ★★★★☆ 8.9 | ✓ ~2–3 Hours | Git diff → interactive module dependency diagram |
| **08. SocraticLab** | EdTech / Negotiation | ★★★★☆ 8.8 | ✓ ~3–4 Hours | Branching AI dialogue with live persuasion radar |
| **09. DataStory** | BI & Analytics | ★★★★☆ 9.1 | ✓ ~2 Hours | Ugly raw CSV → 3 interactive charts & executive deck |

**Official Devpost Learn Hackathon Skill Pack:**

```
npx skills add challengepost/learn-ai-basics --all -y
```

6-Step Flow: `1-start` • `2-scope` • `3-prd` • `4-spec` • `5-build` • `6-ship`